

Chapter 1 _ Introduction to Building AI applications with Foundation Model

Language model = Text-completion

The basic unit of a language model is a **token**.

A token can be a character, a part of a word, depending on the model.

Tokenization is the process of breaking the text into tokens. The tokenization method and vocabulary size are decided by the model developers.

Why do language models use tokens as their unit instead of words or characters?

- Compared to characters, tokens allow the model to break words into meaningful components. For example: "Cooking" can be broken into "Cook" and "ing", with both components carrying some meaning of the original word
- Because there are fewer unique tokens than unique words, this reduces the model's vocabulary size, making the model more efficient
- Token also helps the model to process unknown words, for instance, a made-up word like "chatgpting" could be split to "chatgpt" and "ing", helping the model understand its structure. Tokens balance having fewer units than words while retaining more meaning than individual characters

Language Model:

- **Masked Language model:** trained to predict the missing tokens anywhere in a sequence, using the context from both before and after the missing tokens. Use case: code debugging...
- **Autoregressive Language model:** trained to predict the next token in a sequence, using only the preceding tokens. AKA it predicts what comes next. Use case: write, text generation...

A model can generate open-ended output is called generative, hence, generative AI

Supervision = the process of training an AI algorithm using labeled data, which can be expensive and slow to obtain.

Self-supervision = helps overcome the labelling bottleneck to create larger datasets for models to learn from, efficiently allowing the models to scale up.

Self-supervised learning = language models can learn from text sequences without requiring any labelling.

The model's size is typically measured by the number of parameters.

A parameter is a variation within an ML model that is updated through the training process.

In general, the more parameters the model has, the greater its capacity to learn desired behaviors.

and mark the beginning and the end of a sequence. These markers are essential for a language model to work with multiple sequences. Each marker is typically treated as one special token by the model. The end-of-sequence marker is especially important as it helps language models know when to end their responses.

AI research was divided by data modalities: NLP, CNN, CV,...

The model that can work with more than one data modality is called a **multimodal model**

AI engineering refers to the process of building applications on top of the foundation models.

Enterprise Generative AI can be categorized into:

- Customer experience
- Employee productivity
- process optimization
- programming
- data analysis
- Design, creative concept
- Marketing....

The role of AI

- **Critical or complementary:** the more critical the AI is to the application, the more accurate and reliable the AI part has to be.

- **Reactive or Proactive:**
 - The reactive feature shows its response to the user's query or requests. Example: chatbot
 - The proactive feature shows its response when there is an opportunity for it. Example: Traffic, Google map, (turn left, turn right....)
- **Dynamic or static:**
 - Dynamic is updated continually with user feedback,
 - static feature is updated periodically

Involving humans in the AI's decision-making process is called **Human-in-the-loop**

Hot-take: If you build your product on top of a model trained using other people's data, can you be certain that your product's IP will always belong to you? Many IP-heavy companies in the gaming industry hesitate to use AI for fear of losing their IP's later on.

Three-layer AI stack

- **Application Development:** Involve providing the model with good prompts and necessary context. This layer requires rigorous evaluation. A good application also demands a

good interface.

- Model Development: This layer provides tooling for developing models, including frameworks for modeling, training, finetuning, and inference optimization. This layer also contains dataset engineering and rigorous evaluation.

- Infrastructure: Include tooling for model serving, managing data and compute, and monitoring.

AI engineering focuses less on modeling and training, and more on model adaptation. There is more pressure for efficient training and inference optimization.

Model Adaptation

- Prompt-based technique = include the prompt engineering, adapt a model without updating the model weights.
- Finetuning = requires updating model weights. We can adapt the model by making changes to the model itself.

Developing an AI application

- **Model Development:** modeling and training, dataset engineering, and inference optimization
 - **Model and Training** = the process of coming up with the model architecture, training it, and finetuning it.
 - Training always involves changing model weights, but not all changes to model weights constitute training. For example: quantization
 - Pre-training = training the model from scratch, the weights are randomly initialized
 - Finetuning = continuing to train a previously trained model-- the model weights are obtained from the previous training process.
 - Post-training = Finetuning and Post-training are the same, and are used interchangeably used.
 - **Datasets Engineering:** curating, generating, and annotating the data needed for training and adapting AI models.
 - **Inference Optimization:** making models faster and cheaper. One challenge, Autoregressive = tokens are generated sequentially. if it takes 10 ms for a model to generate a token, it will take a second to generate an output of 100 tokens, and even more for longer outputs.

Category	Building with traditional ML	Building with foundation models
Modeling and training	ML knowledge is required for training a model from scratch	ML knowledge is a nice-to-have, not a must-have ^a
Dataset engineering	More about feature engineering, especially with tabular data	Less about feature engineering and more about data deduplication, tokenization, context retrieval, and quality control
Inference optimization	Important	Even more important

- **Application Development:** Consists of evaluating, prompt engineering, and AI interface
 - **Evaluation:** mitigating risks and uncovering the opportunities to improve
 - **Prompt Engineering and context construction:** getting the model to behave and express the desirable behaviors from the input alone, without changing the model weight.
 - **AI Interface:** creating an interface for the end user to interact with the AI application.